

Technische Hochschule Ingolstadt
Fakultät Elektro- und Informationstechnik

Sensor Network Technologies and Sensor Fusion

Sensor Uncertainty Modeling

Bayes Rule

Nov 6, 2025

Welcome to the lecture!

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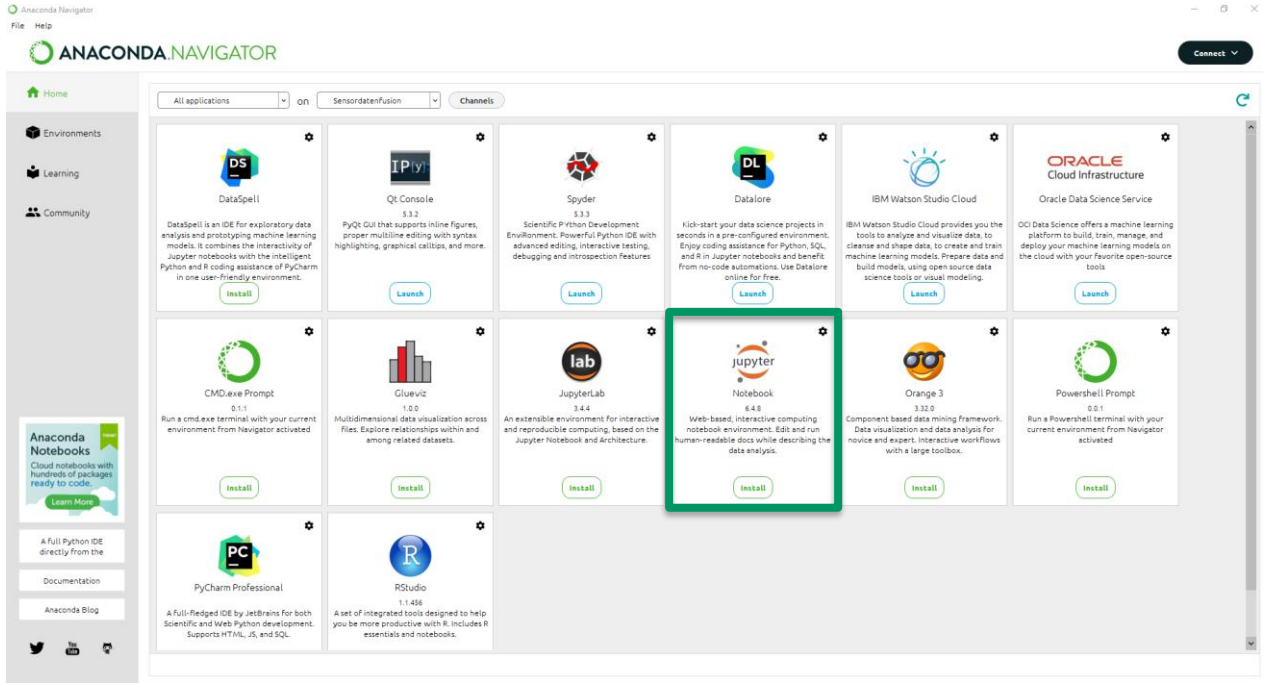
AIMotion, AI and Sensor Data Fusion

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- **Please register for the lecture on Moodle!**
  - Link: https://moodle.thi.de/course/view.php?id=10580
  - Key: **SensNT+DF#WS25**
- **News on Moodle:**
  - Slides for the lecture (upcoming lectures marked as WIP and are subject to change!)
  - Formulary, Exercises (Solutions provided after discussion in the lecture)
  - Exercises are optional, one task / sheet per week, but are highly recommended.
    They can be handed in for correction.
  - I will *try* to publish exercises before Wednesday‘s lecture so that initial questions can be discussed on Thursday.
- **Exam: Written exam, 90 minutes**
- **Consultation hours: Please arrange via e-mail to bjoern.giesler@thi.de.**
- **Please also don‘t hesitate to send questions, lecture feedback, change requests, etc.!**

Software: Python/Anaconda

- For the ML examples we will use today, will use the Jupyter Notebook environment.
- It is recommended you install the Anaconda Python environment and distribution, that installs Jupyter as one of its environments.
- You can find it at https://www.anaconda.com/.

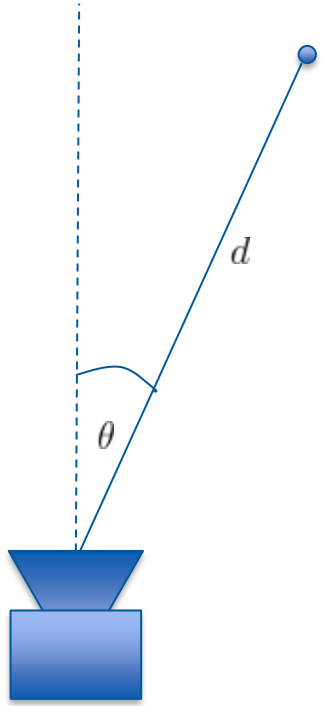


Sensor Modeling

Putting it all together – Sensor Modelling Example

- **Let‘s assume we have a sensor that delivers distance over azimuth.**
- **We take 1000 measurements at various angles and distances and find:**
  - all measurements witthin $\pm 3^\circ$ across the whole field of view
  - all measurements within $\pm 1\%$ depending on distance.

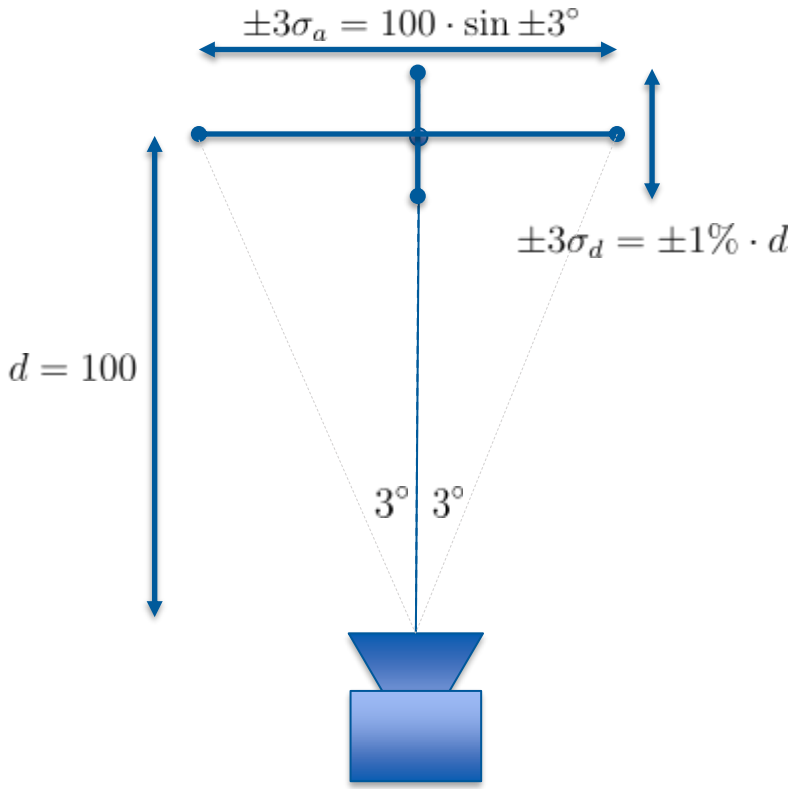
- **What is the covariance matrix of an object measured at 20° of azimuth, 100m of distance?**



- We start with an unrotated normal distribution of $\Sigma_0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.
- Since we empirically measured all measurements within $\pm 1\%$ of distance, we can assume $3\sigma_d = 100 \cdot 1\%$

  or $\sigma_d = \frac{1}{3}$ meters.
- Since we empirically measured all measurements with $\pm 3^\circ$ of azimuth, we can assume $3\sigma_a = 100 \cdot \sin 3^\circ$

  or $\sigma_a \approx \frac{5}{3}$ meters.
- So our unrotated data is scaled by $\begin{bmatrix} \frac{5}{3} & 0 \\ 0 & \frac{1}{3} \end{bmatrix}$.



- **We see the object at 20° of azimuth, so we need to rotate by**

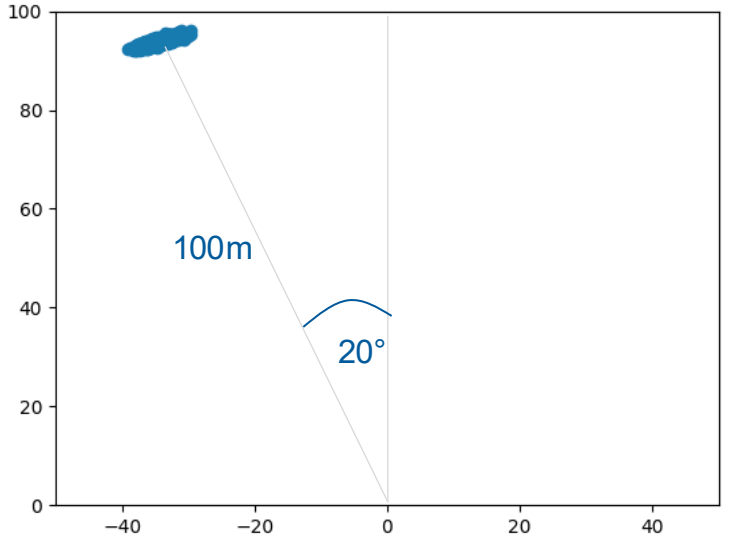
$$\begin{bmatrix} \cos 20° & -\sin 20° \\ \sin 20° & \cos 20° \end{bmatrix} \approx \begin{bmatrix} 0.94 & -0.342 \\ 0.342 & 0.94 \end{bmatrix}$$

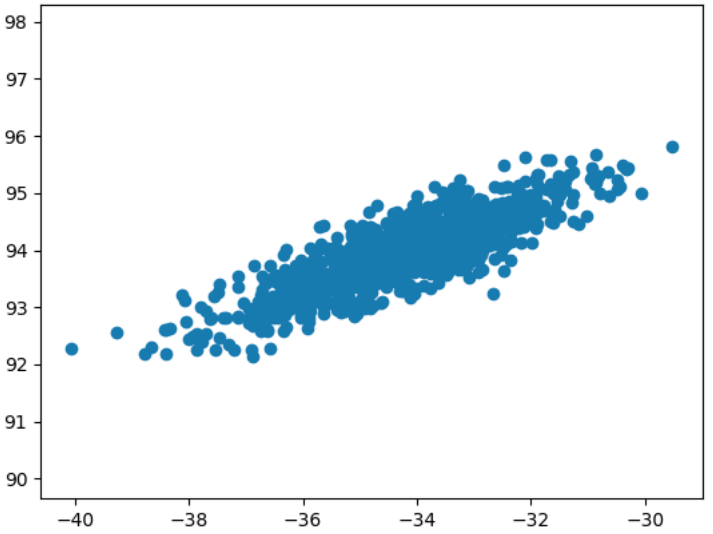
- **For a data transform of** $T = RS = \begin{bmatrix} \cos 20° & -\sin 20° \\ \sin 20° & \cos 20° \end{bmatrix} \begin{bmatrix} \frac{5}{3} & 0 \\ 0 & \frac{1}{3} \end{bmatrix}$

we obtain a covariance matrix of

$$\Sigma' = RS(SR)^T = \begin{bmatrix} \cos 20° & -\sin 20° \\ \sin 20° & \cos 20° \end{bmatrix} \begin{bmatrix} \frac{5}{3} & 0 \\ 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} \frac{5}{3} & 0 \\ 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} \cos 20° & \sin 20° \\ -\sin 20° & \cos 20° \end{bmatrix}$$

and can use this to generate a scatter plot of the multivariate normal distribution at $\vec{\mu} = \begin{bmatrix} 100 \sin \theta \\ 100 \cos \theta \end{bmatrix} \approx \begin{bmatrix} 34.2 \\ 93.97 \end{bmatrix}$ **.**





Multivariate Distributions

- Usually, a system is made of multiple states and multiple measurements
- Therefore, we consider vector-valued random variables $x \rightarrow \boldsymbol{x}$
- Then the variance becomes the covariance

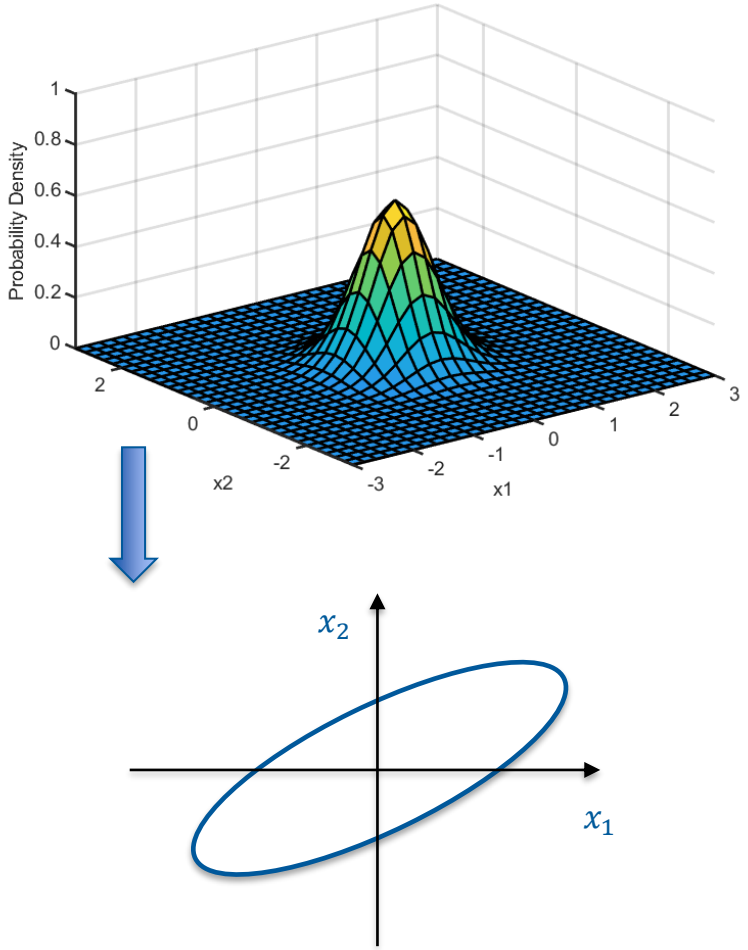
$$\mathrm{Var}[x] = E[(x - E[x])^2]$$

$$\Rightarrow \boldsymbol{P}_{XX} = \mathrm{Cov}\,(\boldsymbol{x}, \boldsymbol{x}) = \boldsymbol{E}\left[(\boldsymbol{x} - \boldsymbol{E}\,[\boldsymbol{x}]) \cdot (\boldsymbol{x} - \boldsymbol{E}\,[\boldsymbol{x}])^{\mathrm{T}}\right]$$

Multivariate Gaussian Distribution

- Notation $\boldsymbol{x} \sim N(\boldsymbol{\mu}, \boldsymbol{P}_{XX})$
- Probability density function

$$p_X(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n \det \boldsymbol{P}_{XX}}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}} \boldsymbol{P}_{XX}^{-1} (\mathbf{x}-\boldsymbol{\mu})}$$



Two-dimensional Gaussian distribution

Gaussian Distributions: Summary

- Affine-Transformation: If $p(\boldsymbol{x})$ is Gaussian, $p(\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b})$ is also Gaussian
- Marginalization: If $p(\boldsymbol{x}, \boldsymbol{y})$ is Gaussian, $p(\boldsymbol{x})$ is also Gaussian
- Conditional distributions: If $p(\boldsymbol{x}, \boldsymbol{y})$ is Gaussian, $p(\boldsymbol{x}|\boldsymbol{y})$ is also Gaussian
- Product: If $p(\boldsymbol{x})$, $p(\boldsymbol{y})$ is Gaussian, $p(\boldsymbol{x}) \cdot p(\boldsymbol{y})$ is proportional to a Gauss-Distribution

These characteristics simplify computations involving Gaussian distributions. Most other distributions do not posses such advantageous characteristics.

Stochastically independent Random Variables, Conditional Probabilities and Bayes Rule

Stochastical Idependency

- If two events A and B are *stochastically independent*, their joint probability of occurance is

$$P(A \cap B) = P(A \text{ and } B) = P(A) \cdot P(B)$$

- If two random variables are stochastically independent, their joint probability density function is

$$p(x, y) = p(x) \cdot p(y)$$

where $p(x) = p_X(x)$

Conditional Probability – Joint Probability for *stochastically dependent* variables

- Probability $P(A|B)$, that event A occurs *conditioned on the occurance* of B
- The probability of an event of occurance of A and B is $\mathrm{P(A \cap B)} = P(A|B) \cdot P(B)$, so the condional probability is

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$ for stochastically independent events $$P(A|B) = \frac{P(A) \cdot P(B)}{P(B)} = P(A)$$

Conditional Probabilities and Marginalization

- For two random variables X and Y the conditional probability density function is obtained as

$$p(x|y) = \frac{p(x,y)}{p(y)}$$

- **Marginalization:**
  - Conditional probability allows us to create the probability distribution function for a single variable from that of two independent ones.
  - We can do that for the continuous case by computing the probability density function of X integrating over all realizations of Y.

$$p(x) = \int_{-\infty}^{\infty} p(x,y)dy = \int_{-\infty}^{\infty} p(x|y)p(y)dy$$

  - Or for the discrete case

$$P(X) = \sum P(x|y)P(y)$$

Bayes Rule

- Based on the joint probability of two events A and B we obtain the following rule for conditional probabilities

$$P(A \cap B) = P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$

- From the above equation we obtain **Bayes Rule**

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Example 1:

$$P(\{6\} \cap \{2,4,6\}) =$$
$$P(\{6\}|\{2,4,6\}) \cdot P(\{2,4,6\}) = P(\{2,4,6\}|\{6\}) \cdot P(\{6\}) \Rightarrow$$
$$P(\{6\}|\{2,4,6\}) = \frac{P(\{2,4,6\}|\{6\}) \cdot P(\{6\})}{P(\{2,4,6\})} = \frac{1 \cdot 1/6}{1/2} = \frac{1}{3}$$

Example 2:

- What is the probability, that there is fire if we detect smoke?
- To answer this question, the following information is provided:
  - Probability of fire 1% => $P(fire)$
  - Probability of detecting smoke 10% => $P(smoke)$
  - 90% of all fires produce smoke => $P(smoke|fire)$

Solution

- Applyíng Bayes rule, we obtain

$$P(\text{Fire}|\text{Smoke}) = \frac{P(\text{Smoke}|\text{Fire}) \cdot P(\text{Fire})}{P(\text{Smoke})}$$
$$= \frac{0.9 \cdot 0.01}{0.1} = 0.09$$

Example 3:

- We analyze the punctuality of trains, departing from Ingolstadt
- The departing trains have the following destinations:
  - D1 Berlin 50% of all trains go there, so the likelihood of any train going to Berlin is 0.5.
  - D2 Cologne 30% of all trains go here
  - D3 Munich 20% of all trains go here
- The probability that a train will arrive delayed at the destination is:
  - D1 Berlin 40%
  - D2 Cologne 35%
  - D3 Munich 25%
- Now choosing randomly among the delayed trains at the destinations what is the probability that its destination is Berlin?

Example 2:

- We analyze the punctuality of trains, departing from Ingolstadt
- The departing trains have the following destinations:
  - D1 Berlin 50% => $P(D1)$
  - D2 Cologne 30% => $P(D2)$
  - D3 Munich 20% => $P(D3)$
- The probability, that a train will arrive delayed at the destination is:
  - D1 Berlin 40% => $P(V|D1)$
  - D2 Cologne 35% => $P(V|D2)$
  - D3 Munich 25% => $P(V|D3)$
- Now choosing randomly among the delayed trains at the destinations what is the probability that its destination is Berlin? => **We are looking for** $P(D1|V)$

Example 2:

- We analyze the punctuality of trains, departing from Ingolstadt
- Now choosing randomly among the delayed trains at the destinations what is the probability that its destination is Berlin? => **We are looking for** $P(D1|V)$
- We compute

$$P(D1|V) = \frac{P(V|D1) \cdot P(D1)}{P(V)}$$

- The probability of a delayed arrival is obtained by means of

$$\begin{aligned} P(V) &= P(V|D1) \cdot P(D1) + P(V|D2) \cdot P(D2) + P(V|D3) \cdot P(D3) \\ &= 0.5 \cdot 0.4 + 0.3 \cdot 0.35 + 0.2 \cdot 0.25 = 0.355 \end{aligned}$$

- So we finally obtain

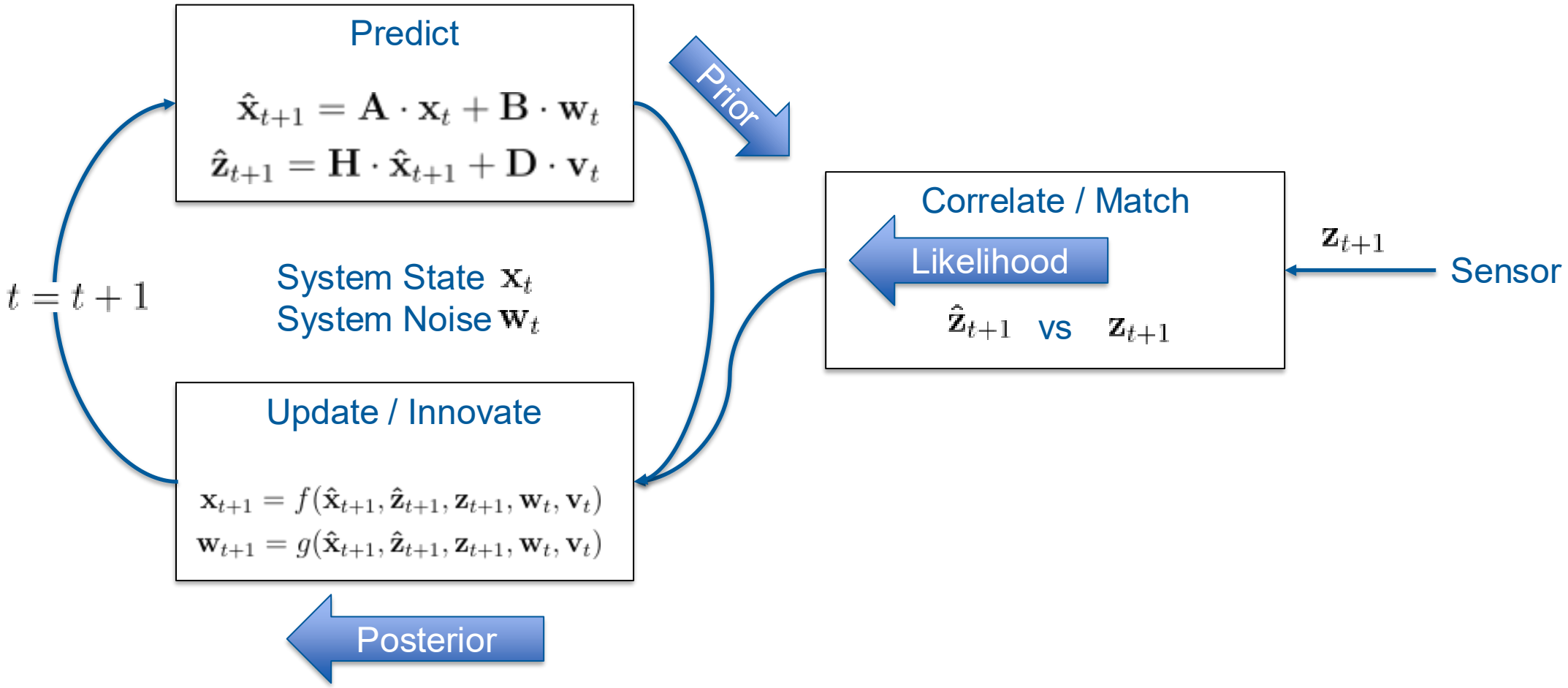
$$P(D1|V) = \frac{0.5 \cdot 0.4}{0.355} = 0.5634$$

Bayes Rule - Interpretation

- Update of a hypothesis H based on data/observations/evidence D

$$P(H|D) = \frac{P(D|H) \cdot P(H)}{P(D)}$$

- Interpretation
  - **Posterior:** Probability of a hypothesis after observing the data: $P(H|D)$
  - **Prior:** Probability of the hypothesis before observing the data: $P(H)$
  - **Likelihood:** Probability of the data under the hypothesis: $P(D|H)$
  - Normalization factor $P(D)$


Predict
$\hat{\mathbf{x}}_{t+1} = \mathbf{A} \cdot \mathbf{x}_t + \mathbf{B} \cdot \mathbf{w}_t$
$\hat{\mathbf{z}}_{t+1} = \mathbf{H} \cdot \hat{\mathbf{x}}_{t+1} + \mathbf{D} \cdot \mathbf{v}_t$
Prior
Correlate / Match
Likelihood
$\hat{\mathbf{z}}_{t+1}$ vs $\mathbf{z}_{t+1}$
$\mathbf{z}_{t+1}$
Sensor
$t = t + 1$
System State $\mathbf{x}_t$
System Noise $\mathbf{w}_t$
Update / Innovate
$\mathbf{x}_{t+1} = f(\hat{\mathbf{x}}_{t+1}, \hat{\mathbf{z}}_{t+1}, \mathbf{z}_{t+1}, \mathbf{w}_t, \mathbf{v}_t)$
$\mathbf{w}_{t+1} = g(\hat{\mathbf{x}}_{t+1}, \hat{\mathbf{z}}_{t+1}, \mathbf{z}_{t+1}, \mathbf{w}_t, \mathbf{v}_t)$
Posterior

Bayes Rule applied to Probability Density Functions

- Probability density functions are better suited to express uncertainties of hypotheses and prior information. For density functions we have that

$$p(\theta|D) = \frac{p(D|\theta) \cdot p(\theta)}{p(D)} \quad \Rightarrow \quad \text{Posterior} \propto \text{Likelihood} \cdot \text{Prior}$$

- Interpretation
  - **Posterior:** Probability density function of the hypothesis θ after observing the data $p(\theta|D)$
  - **Prior:** Probability density function of the hypothesis before observing the data $P(\theta)$
  - **Likelihood:** Probability density function of the data under the hypothesis $P(D|\theta)$
  - Normalization factor $p(D)$

$$p(D) = \int p(D|\theta) \cdot p(\theta) d\theta$$

quite often, the knowledge of $p(D)$ is not required, since we are only interested in the location of $p(\theta|D)$



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Thank you very much!